Govt. M.A.O. Graduate College, Lahore.

MIDTERM EXAMS (May 2025)

Department/Discipline: Physics
Paper: Quantitative Reasoning (II)
Course code: GQR-201
Max. Marks: 25
SEMESTER: 8th (EVENING)
Total Time: 1:00 Hour

Student's Name: ____________________ College Roll No. ____________ PU Roll No.____________

Q.1 Write the converse, inverse and contrapositive of the following conditional:

I. $\neg q \rightarrow \neg p$

II. Prove that $p \vee (\neg p \wedge \neg q) \vee (p \wedge q) = p \vee (\neg p \wedge \neg q)$

Q.2 Compute the slope of the line segment connecting the two points

I. (5 , 10) and (6 , 12)

II. Rewrite equation in slope-intercept form and determine the slope and y-intercept of the following: $5y - 2x + 10 = 3x - 4y + 5$

Q.3 Differentiate the following function: $y = \left(\frac{2x^2+3}{5x+10}\right)^2$

Q.4 Find velocity and acceleration at t= 0,1 ,2. For $s = 2t^2 + 4t + 10$

Q.5 If A= {2,3,7,9,11,13} and B= {1,2,4,6,13,15} find union and intersection by using Venn Diagram.

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